Question	Answer	Marks	Partial Marks
1(a)	4	**1**	
1(b)	9	**1**	
2	1, 2, 4, 17, 34, 68	**2**	**B1** for 5 correct and no extras or 6 correct and 1 extra
3	$4 \times (6 - 2) + 1 = 17$	**1**	
4	$\frac{1}{4}$ or 0.25	**1**	
5(a)	576	**1**	
5(b)	13	**1**	
6	6.8	**1**	
7	381.42 cao	**1**	
8	1098 cao	**2**	**M1** for $\frac{18}{2}(7 \times 18 - 4)$ or better
9	Correct ruled triangle with construction arcs	**2**	**B1** for correct triangle with incorrect/no arcs or for 2 correct arcs
10	12.66 cao	**2**	**B1** for 12.657 to 12.658 or 12.7 or **M1** for $\frac{924}{73}$ If 0 scored, **SC1** for *their* 12.657… seen and correctly rounded to 2 dp
11(a)	Tangent	**1**	
11(b)	Chord	**1**	
11(c)	3.72 or 3.720 to 3.721…	**2**	**M1** for $[r^2 =]\ \frac{43.5}{\pi}$ or better
11(d)	201 or 201.0 to 201.1	**3**	**B2** for 20.1 or 20.10 to 20.11 [cm] or 64π or **M1** for $6.4 \times \pi$ oe or **B1** for *their* 20.1 [cm] seen and correctly converted to mm
12(a)	34	**1**	

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Question	Answer	Marks	Partial Marks
12(b)	24 66	**2**	**B1** for 1 correct
13(a)	18	**1**	
13(b)(i)	1 [hour] 20 [mins]	**2**	**M1** for $\frac{20}{15}$ or 1.333… [hours]
13(b)(ii)	Line from (12 00, 30) to Town D passing through (13 00, 45)	**1**	
13(c)	72	**1**	
13(d)	11.5 or 11.53 to 11.54	**3**	**B1**FT for 4.333… or 4 hours 20 min **M1** for $\frac{50}{3 + \textit{their}\ \textbf{(b)(i)}\ \text{in hours}}$ oe
13(e)(i)	Line from (11 00, 30) to (13 42, 0)	**2**	**B1** for either (11 00, 30) or (13 42, 0) correctly plotted
13(e)(ii)	27	**1**	**FT** *their* line in **(e)(i)**
14(a)(i)	120	**2**	**M1** for $6 \times 4 \times 5$ oe
14(a)(ii)	$2 \times (6 \times 4 + 4 \times 5 + 6 \times 5)$ leading to 148	**M2**	**M1** for 6×4 or 4×5 or 6×5 oe soi
14(a)(iii)	60	**2**	**M1** for $4 \times 6 + 4 \times 5 + 4 \times 4$ oe
14(b)	$6x + 8y - 2$ final answer	**3**	**B2** for 2 correct terms in final answer or for $6x + 8y - 2$ seen and then spoilt or **B1** for either $6x$ or $8y$ or -2 in final answer
15(a)	3.75 or 3.752 to 3.753...	**2**	**M1** for $[r^2 =]\ \frac{177}{4\pi}$ or better
15(b)	221 or 220.8 to 221.5	**2**	**FT** *their* radius **M1** for $\frac{4 \times \pi \times (\textit{their}\ \text{radius})^3}{3}$ oe
16(a)	73 500 157 500	**3**	**B2** for either correct or **M1** for 8 parts = 84 000 soi
16(b)(i)	6720	**1**	
16(b)(ii)	63	**2**	**M1** for $\frac{56000 - \textit{their}\ 6720 - 14000}{56000}$ [× 100] oe

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Question	Answer	Marks	Partial Marks
16(b)(iii)	15 400 or 15 390 or 15 393 or 15 393.[16…]	**2**	**M1** for $14\,000\left(1+\frac{2.4}{100}\right)^4$ oe
17(a)	144	**1**	
17(b)	24	**2**	**M1** for 2, 2, 2, 3, 5 or better or a correct factor tree or list/table for 120
17(c)	720	**1**	
18(a)	[a =] 13 [b =] 19 [c =] 25	**2**	**B1** for 2 terms correct or for 2 terms with a difference of 6
18(b)(i)	$7n - 3$ oe final answer	**2**	**B1** for $7n + j$ or $kn - 3$ $k \neq 0$ as final answer or for $7n - 3$ oe seen then then further incorrect working
18(b)(ii)	*their* **(b)(i)** = 361 oe	**M1**	
	Correct solution leading to 52	**A1**	
19	41.25	**3**	**M2** for $\frac{12 \times 43.25 + 16 \times 39.75}{28}$ oe or **M1** for 12×43.25 or 16×39.75 oe
20	1.5 oe	**2**	**B1** for figs 15 or **M1** for $\frac{5.4 \times 1000}{60 \times 60}$ oe
21(a)	Angle in a semicircle is 90	**1**	
21(b)	7.31 or 7.310 to 7.311	**3**	**M2** for $\frac{6.2}{\cos 32}$ oe or **M1** for $\frac{6.2}{PQ}$ oe
22	5.4[0] or 5.395…	**3**	**M2** for $5.6^2 - 1.5^2$ oe or **M1** for $5.6^2 = [\ldots]^2 + 1.5^2$ oe

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